# Practice workbook · Write or complete the respiration equations

Word and balanced symbol equations for aerobic respiration and the word equation for anaerobic respiration in humans.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-RESPIRATION-EQUATION

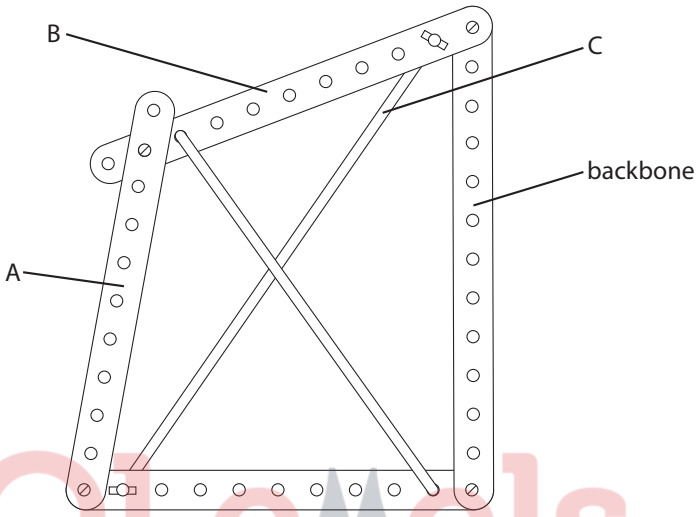
# 1. Question 5 · part b · 3 marks

June 2025 Paper 1

Worked-example retry · Attempt part b only.

**5** (a) The diagram shows a model that can be used to demonstrate the process of breathing.

The part that represents the backbone has been labelled.




(i) State which parts of the breathing system are represented by A, B and C.

**(3)**

A .....

B .....

C .....

(ii) State three other structures, not shown in the diagram, that are needed for breathing in the body.

**(3)**

1 .....

2 .....

3 .....

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# 1. Question 5 · part b · 3 marks

June 2025 Paper 1

Worked-example retry · Attempt part b only.

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(iii) Describe how the structures present in the thorax are used to allow a person to inhale.

**(6)**

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(b) Give the balanced chemical symbol equation for the process of aerobic respiration.

**(3)**

..... + ..... → ..... + ..... + .....

**(Total for Question 5 = 15 marks)**

## 1. Question 5 · part b · 3 marks · Working and answer

Working and answer

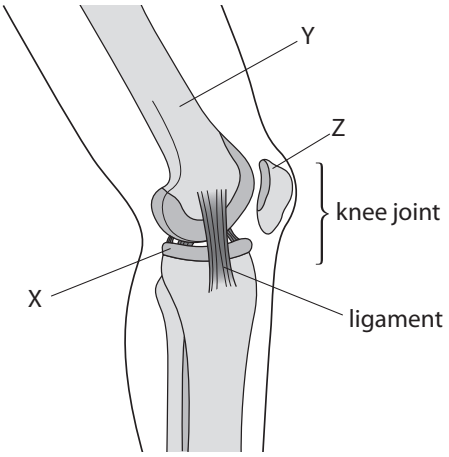
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## 2. Question 1 · part b · 2 marks

June 2019 Paper 1

Further practice · Attempt part b only.

**1** (a) The diagram shows structures in part of a human leg.




(i) The box lists structures in the human leg.

| | | | |
|---|---|---|---|
| cartilage | femur | kneecap | muscle |
| fibula | tendon | tibia | |

Use words from the box to name structures X, Y, and Z.

**(3)**

X .....

Y .....

Z .....

(ii) What is the function of ligaments in the knee joint?

**(1)**

- ☒ **A** ligaments attach muscles to the leg bones
- ☒ **B** ligaments hold the leg bones together
- ☒ **C** ligaments move the leg bones
- ☒ **D** ligaments stop the leg bones rubbing together

(iii) The type of joint found in the knee and the elbow is the same.

Give the name of this type of joint.

**(1)**

.....

**2**

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## 2. Question 1 · part b · 2 marks

June 2019 Paper 1

Further practice · Attempt part b only.

(b) Aerobic respiration releases energy for muscles to contract.

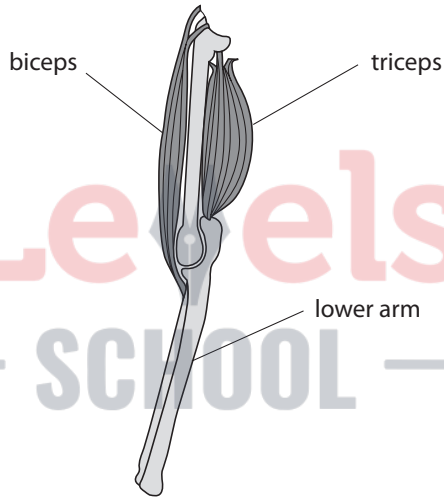
Complete the word equation for aerobic respiration.

**(2)**

glucose + ..... → ..... + water

(c) Diagram 1 shows a model of the human arm, with the lower arm extended.

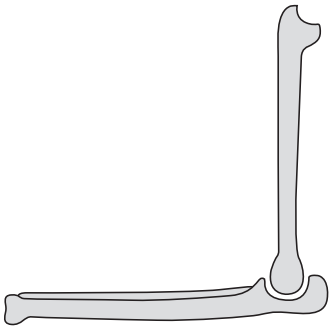
**Diagram 1**




Complete diagram 2 to show how the muscles change when the lower arm is raised.

**(2)**

**Diagram 2**



**(Total for Question 1 = 9 marks)**

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## 2. Question 1 · part b · 2 marks · Working and answer

Working and answer

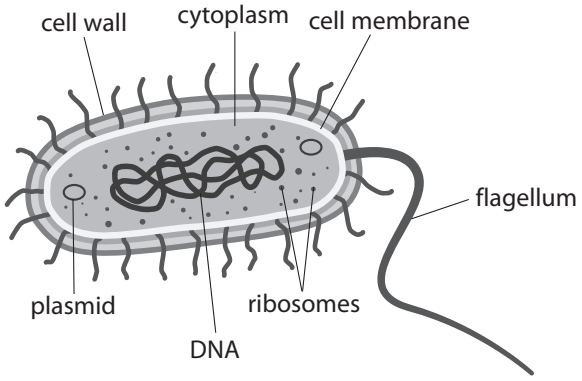
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# 3. Question 3 · part b.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

**3** (a) The diagram shows a type of bacterial cell.




(i) The boxes give some structures found in the bacterial cell and some functions of cell structures.

Draw one straight line from each structure to its correct function.

**(3)**

| Structure | Function |
|---|---|
| cell membrane | protects against pathogens |
| | helps the cell to move |
| ribosomes | makes proteins |
| | holds the genetic code |
| flagellum | controls which substances enter and exit the cell |

(ii) State two structural differences between the bacterial cell and a sperm cell.

**(2)**

1 .....

.....

2 .....

.....

**8**

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# 3. Question 3 · part b.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

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(b) Many bacteria absorb oxygen and nutrients from their environment. Oxygen and nutrients are used by bacteria in aerobic respiration.

(i) Which process transports oxygen into bacterial cells?

(1)

- ☒ **A** active transport
- ☒ **B** diffusion
- ☒ **C** excretion
- ☒ **D** osmosis

(ii) Complete the word equation for aerobic respiration.

(2)

oxygen + ..... → ..... + water

(iii) Explain why respiration is important for bacteria.

(2)

.....

.....

.....

.....

**(Total for Question 3 = 10 marks)**

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## 3. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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# 4. Question 2 · part a · 4 marks

June 2021 Paper 2

Further practice · Attempt part a only.

**2** Aerobic respiration is the process that releases energy for use in the body.

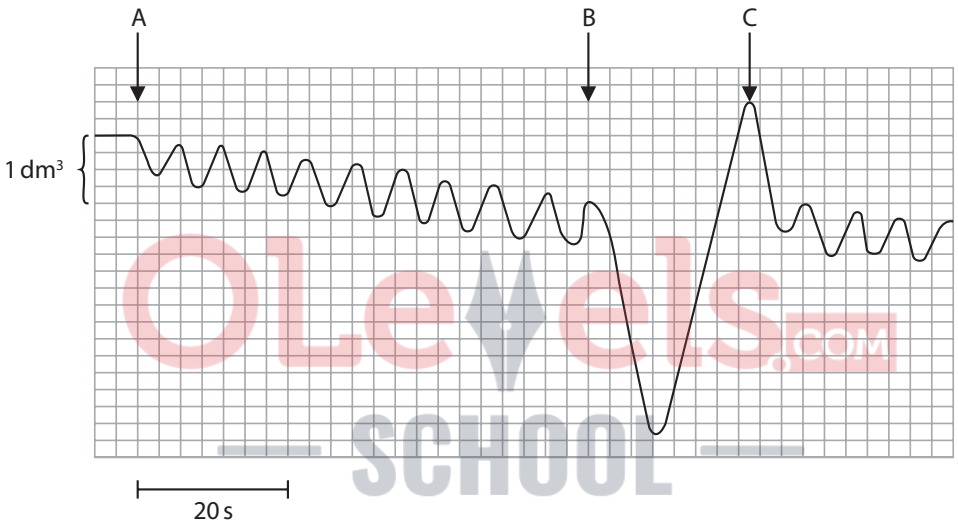
(a) Give the word equation for the process of aerobic respiration.

**(4)**

..... + ..... → ..... + ..... + energy

(b) A spirometer is used to measure the volumes of air breathed in and the volumes of air breathed out.

The spirometer trace shows the readings taken of a person at rest.




(i) Calculate the rate of breathing between points A and B shown on the trace.

**(2)**

rate = ..... breaths per minute

(ii) Calculate the difference in the volume of air breathed in and the volume of air breathed out between points B and C.

**(3)**

difference in volume = ..... $dm^3$

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## 4. Question 2 · part a · 4 marks · Working and answer

Working and answer

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# 5. Question 5 · part a.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.i only.

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**5** Some athletes train at high altitudes to improve their performance.

Performance improves if more energy is released from aerobic respiration.

(a) (i) Complete the balanced symbol equation for aerobic respiration.

**(2)**

..... + $6O_2$ → ..... + $6CO_2$

(ii) Aerobic respiration produces ATP.

Describe how energy is released from ATP during exercise.

**(2)**

.....

.....

.....

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## 5. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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## 6. Question 4 · part a.i · 2 marks

June 2023 Paper 2

Further practice · Attempt part a.i only.

**4** (a) (i) Complete the word equation for one type of respiration.

**(2)**

oxygen + ..... → ..... + .....

(ii) State the type of respiration shown by the equation.

**(1)**

.....

(iii) In which part of the cell does this type of respiration occur?

**(1)**

- ☒ **A** endoplasmic reticulum
- ☒ **B** mitochondrion
- ☒ **C** nucleus
- ☒ **D** ribosome

(iv) Name the molecule used to transfer energy in a cell.

**(1)**

.....

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## 6. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

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## 7. Question 4 · part a.ii · 1 mark

June 2023 Paper 2

Further practice · Attempt part a.ii only.

**4** (a) (i) Complete the word equation for one type of respiration.

**(2)**

oxygen + ..... → ..... + .....

(ii) State the type of respiration shown by the equation.

**(1)**

.....

(iii) In which part of the cell does this type of respiration occur?

**(1)**

- ☒ **A** endoplasmic reticulum
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(iv) Name the molecule used to transfer energy in a cell.

**(1)**

.....

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## 7. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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## 8. Question 7 · part a.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part a.i only.

**7** (a) Aerobic and anaerobic respiration both take place during exercise.

(i) Give the word equation for anaerobic respiration.

**(1)**

.....

(ii) Complete and balance the chemical symbol equation for aerobic respiration.

**(2)**

$C_6H_{12}O_6$ + ..... ⟶ ..... + ..... $CO_2$

(iii) A student investigates the rate of carbon dioxide production during exercise and at rest.

This is the equipment they use.

- four pieces of glass tubing
- two boiling tubes
- two rubber bungs
- mouthpiece
- limewater

Draw a diagram to show how the student should set up the equipment to carry out the investigation.

**(2)**

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**20**

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## 8. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

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## 9. Question 7 · part a.ii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part a.ii only.

**7** (a) Aerobic and anaerobic respiration both take place during exercise.

(i) Give the word equation for anaerobic respiration.

**(1)**

.....

(ii) Complete and balance the chemical symbol equation for aerobic respiration.

**(2)**

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- limewater

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**(2)**

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## 9. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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